TRAN NHAT PHI AI Engineer

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About Me | I'm an **Information Technology graduate** with **over 2+ years of hands-on experience** in **software and AI development**. Through my studies, internships, and real-world projects, I've built strong **technical skills**, a solid **problem-solving mindset**, and a passion for creating **practical, scalable solutions**. I'm eager to **apply my knowledge and continue learning** while contributing to **innovative and impactful projects**. I look forward to the opportunity to bring my **skills**, **experience**, and **enthusiasm** to your team!

Experience

LNCGLOBAL CONSULTANT JOINT STOCK COMPANY **6/2026 - present**

Pre-Middle AI Engineer

Technologies: Docker, Docker Compose, Kubernetes (k9s), RabbitMQ, FastAPI, Next.js, Python, MySQL, SQLAlchemy, MinIO (S3), Tesseract OCR, OpenCV, DeepSeek LLM API, Vector Database, Neo4j, Graphiti, Ontology RAG, Ragas

Responsibilities:

- Designed and built an **Ontology-based RAG** system modeling **immigration program entities and relationships** (eligibility criteria, visa categories, processing requirements), improving retrieval precision for multi-condition eligibility questions by **28%** compared to plain vector search.
- Prototyped **Graphiti** on top of **Neo4j** to build a **temporal knowledge graph**, tracking how **immigration policy provisions and effective dates** evolve over time across programs (EB3, NIW, LMIA), reducing outdated-answer errors by **20%** on time-sensitive policy queries.
- Implemented **hybrid retrieval** (vector + BM25 + graph traversal), increasing answer relevance (measured via Ragas context precision) from **0.71 to 0.89** on the internal evaluation set.
- Built a **document preprocessing pipeline** for **immigration policy and legal case files** (PDF/DOCX), processing over **500+ documents** and preserving hierarchical structure (chapters/clauses) for accurate downstream retrieval.
- Evaluated and monitored RAG pipeline quality using **Ragas** metrics (faithfulness, answer relevance, context precision) across **X test queries/week**, iterating on chunking strategy to cut hallucination rate by **15%**.
- Developed **REST APIs** with **FastAPI** exposing the RAG pipeline to internal case-management tools, using **RabbitMQ**-based task queuing to reliably handle bursts of concurrent requests, eliminating timeout errors during peak document-processing loads.
- Designed and built an internal **document-checklist verification system** for Canada immigration case processing, combining **Next.js** (Server Components) and **FastAPI** with a **Tesseract OCR + OpenCV** preprocessing pipeline and **LLM-based (DeepSeek) classification** against a 29-item document checklist, backed by **MySQL** and **MinIO** (S3-compatible storage).
- Diagnosed and fixed a critical **FastAPI concurrency defect** where a misconfigured async route handler blocked the event loop during file processing, freezing the server for all users — verified fix via live load testing, cutting concurrent-request latency from a **10+ second timeout to ∼50ms**.

VINA INFORMATION TECHNOLOGY COMPANY LIMITED **10/2025 - 5/2026**

Junior Fullstack Developer

Technologies: Frappe, Python, MariaDB/MySQL, React (Ant Design), Elasticsearch, Docker, Kubernetes (k9s), RabbitMQ, REST API, AI Service Integration, LangChain, Vector Database, OCR

Responsibilities:

- Developed ERP modules using **Frappe** (**DocType**, **API**) for order, inventory, manufacturing, booking, and shipping.
- Implemented business logic for **order lifecycle**, **inventory tracking**, and **manufacturing processes**.
- Integrated **React-based UI** into Frappe Desk for complex operational workflows.
- Optimized **order processing logic** to improve system performance and reliability, reducing processing latency by **35%**.
- Integrated **RabbitMQ** for asynchronous task processing and message queue handling, improving background job reliability and system scalability.
- Integrated **Elasticsearch** for centralized log indexing, search, and filtering, improving debugging and incident investigation speed by **40%** across application services.
- Managed and monitored services on **Kubernetes** using `k9s`, supporting debugging and operational visibility.
- Designed **AI integration flow** for ERP system through **Python AI services**, API gateway, and Frappe backend.

- Developed a Chatbot using **RAG** (**Retrieval-Augmented Generation**) to provide personalized consultation for ERP users, improving information retrieval efficiency by **35%**.
- Developed **AI assistant** integrated with ERP APIs to support natural language queries and operations.
- Implemented **OCR-based data extraction pipeline** for automated document processing into ERP system, achieving **85% extraction accuracy** and reducing manual data entry effort.
- Built recommendation logic for **product and supplier suggestions** based on historical ERP data.
- Built real-time observability dashboards with **Prometheus**, **Grafana**, and **Loki** to monitor API performance, system resource usage, service health, and centralized application logs.

CODEZX COMPANY LIMITED **6/2024 – 4/2025**

AI Research Engineer & Full Stack Developer

Technologies: Python, PaddleOCR, OpenAI API, Java Spring Boot, Angular 17, PostgreSQL, Docker, Jenkins, AWS EC2, AWS S3, AWS RDS, Nginx, Prometheus, Grafana

Responsibilities:

- Developed **AI-powered OCR pipelines** for invoice recognition using **PaddleOCR** and **OpenAI GPT-3.5 Turbo API**, extracting invoice data into structured **JSON outputs** for internal processing systems.
- Optimized **OCR inference**, **prompt engineering**, and image preprocessing workflows, improving invoice data extraction accuracy to **85%** and reducing processing latency by **30%**.
- Built scalable backend services and authentication systems using **Java Spring Boot**, **PostgreSQL**, and **JWT** to support AI processing and internal management workflows.
- Developed invoice visualization and annotation platforms using **Angular 17**, improving validation workflows and reducing manual review effort by **35%** for operations teams.
- Deployed cloud-based services on **AWS EC2**, using **AWS S3** for document storage and **AWS RDS** for database management; configured **Nginx reverse proxy** and **HTTPS** for production deployment.
- Implemented **CI/CD pipelines** using **Docker** and **Jenkins**, automating build and deployment workflows and reducing manual release effort by **40%**.
- Built real-time observability dashboards with **Prometheus** and **Grafana** to monitor API performance, system resource usage, and service health.

AMAZON WEB SERVICES VIETNAM COMPANY LIMITED (INTERN) **9/2023 - 2/2024**

Intern DevOps Engineer

https://trannhatphi.github.io/awsDeployment/

Technologies: Docker, AWS S3, EC2, RDS, Jenkins, GitHub Actions, CloudWatch, Nginx

Responsibilities:

- Containerized and deployed backend and web services using **Docker**, **AWS EC2**, and **S3**, improving deployment consistency across environments.
- Built and maintained **CI/CD pipelines** with **Jenkins** and **GitHub Actions**, reducing manual deployment effort by **40%**.
- Configured **Nginx** as a reverse proxy for web services, supporting request routing, service exposure, and application deployment on cloud infrastructure.
- Monitored **AWS resources** including **EC2**, **S3**, and **RDS** using **CloudWatch metrics and alarms**, improving issue detection speed by **30%**.
- Supported database deployment and connectivity setup using **Amazon RDS** for backend services.
- Documented deployment procedures, infrastructure setup, and system architecture diagrams, reducing onboarding and troubleshooting time by **25%**.

Competitions & Achievements

ICPC 2025: **Qualified for and participated in the ICPC 2025 Asia Regional Contest**. Solved competitive programming problems across **DP, graphs, combinatorics, and algebraic techniques**, reinforcing **teamwork, algorithm design, and high-performance coding skills**.

IMDES2-25: Tung Le, Dat Nguyen, Nhat Le, Phi Tran, "**Comparing Transformer Models for Vietnamese News Classification on VnExpress Data**", published in ***The 25th International Multidisciplinary Engineering, Design, Sciences Symposium (IMDES2-25)***.

GSET 2025: Phi Tran, Nhat Le, Huong Bui, "**Automated Invoice Processing Using OCR and NLP for Multi-format Invoices**", published in ***The 2nd International Conference on Green Solutions and Emerging Technologies for Sustainability (GSETS 2025)***.

OLP Asia 2021: **Participated in the Asia-level Vietnam Student Informatics Olympiad**, practicing **high-performance competitive programming in algorithms, data structures, and optimization**.

Education

Ho Chi Minh City University of Technology **2021 - 2025**

INFORMATION TECHNOLOGY *Language: English & Vietnamese*

GPA: 3.3/4